11th European Conference on Earthquake Engineering © 1998 Balkema, Rotterdam, ISBN 90 5410 982 3

Influence of free field variability on soil-structure interaction

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Keywords: soil-structure interaction, boundary elements method, spectral analysis, spatial variability

ABSTRACT: An integral equation for the influence of spatial variability of the incident free field on seismic soil-structure interaction is proposed. The fundamental solution for the overall domain when the unit load is applied on the structure, is required. It is obtained by means of sub-structuring techniques and boundary integral equations using the Green tensors for homogeneous or horizontally stratified soil media. The effects of a non-stationary modulated random incident field are addressed in terms of the instantaneous power spectral density of the structural response for a given coherency function of the free-field on the traction-free soil surface. For embedded foundations, this coherency function can be computed by a stochastic deconvolution method.

2 INTRODUCTION

The analysis of the random linear vibrations of elastic structures under stationary stochastic loads is classically based on a discretization by the finite element method in order to compute a finite set of eigenmodes of the associated conservative system. The stationary response of the structure is obtained by a standard spectral analysis technique which combines a reduced number of degrees of freedom, namely the modal coordinates on the truncated eigenmodes basis, when the excitation process is projected on this basis. However this approach can not be applied in some given cases, such as vibration analysis in the medium frequency range or coupling problems with infinite media.

Soil-structure interaction problems under random seismic excitations can hardly be computed by the finite element method at reasonable numerical costs. Boundary element methods have proved to be much more efficient in such cases when they are implemented in combination with a sub-structuring technique (Aubry 1986, Clouteau 1990). However they remain limited to particular soil profiles, mainly vertically heterogeneous media. To study the random vibrations of the structure, either the simulation methods (Poirion & Soize 1989) of a set of trajectories of the excitation process, or the standard spectral analysis technique, are used. These approaches require the explicit knowledge of the frequency response matrix of the coupled system, whose dimension is the number of loaded degrees of freedom of the discretized soil-structure interface.

The random load considered in this study is the incident seismic free field for a given site. Recordings of strong motion arrays implemented on several test sites, such as SMART-1 in Taiwan (see for instance Loh & Yeh 1988) or EUROSEISTEST in Greece (Pitilakis et al. 1994), have indicated that the free field may have some lateral variability even over short distances. This variability, apart from the purely deterministic wave passage effect, is strongly incoherent and can be attributed to various complex effects: the geology and stratification of the site, the wave propagation pattern, the wave components depending on the source mechanism, hysteretic damping and geometrical dispersion, diffusion, multiple scattering, etc. The relative importance of these phenomena is clearly dependent on the spatial scale of the analysis and the frequency range considered, however their intrinsic complexity justifies the use of a stochastic model for the incident field. In the instance of large spread foundations, its spatial variability may induce significant differential displacements between two points, and then important additional displacements and efforts in the structure.

In this paper, we first introduce the theoretical formulation used to address the effects of the free field spatial variability on soil-structure interaction. It is based on an integral representation approach. Then the structural response of interest is studied by the spectral analysis technique for a time modulated incident stochastic field described by its transverse coherency function on the ground surface. Thus we derive the frequency response matrix of the coupled system from the integral representation. It simply corresponds in the time domain to a convolution of the free field integrated over the soil-structure interface. As an example, this procedure is applied to the Hualien containment model in Taiwan. When foundations are embedded, the evaluation of the coherency function of the free field in depth is needed and can be computed by a stochastic deconvolution method.

3 NOTATIONS AND HYPOTHESIS

3.1 *Geometry*

The elastic structure under investigation is identified by an open bounded set Ω_b. The infinite visco-elastic soil domain is identified by an open unbounded set Ω_s. The soil part currently occupied by the (possibly embedded) structure foundation is an open bounded set denoted by Ω_{sf}. Γ_{ab} is the traction-free surface of the structure, Γ_{as} is the traction-free surface of the soil and Γ is the interface between the soil and structure domains. We therefore have $\partial\Omega_{sf} = \Gamma_{as} \cup \Gamma$. The overall domain of the model is $\Omega = \Omega_b \cup \Omega_s$. Finally, the radiation surface of the soil $\Gamma_{s\infty}$ is introduced for convenience (Fig. 1).

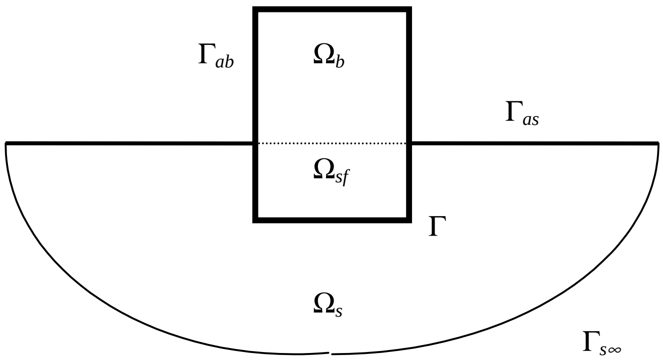

Figure 1. Geometry of the problem and notations.

3.2 *Hypothesis for the free field*

The incident free field, i.e. the seismic field, is denoted $\mathbf{u}_i(\mathbf{x},t)$ in the time domain. It is modeled by a second-order Gaussian non-homogeneous, non-stationary centered stochastic field $\mathbf{U}_i(\mathbf{x},t)$ indexed on $\Gamma\times 3^+$ with values in 3^3. It is therefore completely described by its auto-correlation function defined on $\Gamma\times\Gamma\times 3^+\times 3^+$ with values in $\mathrm{Mat}_3(3,3)$:

$$\mathbf{x},\mathbf{x}',t,t' \mapsto \mathbf{R}_{\mathbf{U}}(\mathbf{x},\mathbf{x}',t,t') = E\{\mathbf{U}_i(\mathbf{x},t) \otimes \mathbf{U}_i(\mathbf{x}',t')\} \quad (1)$$

where E is the mathematical expectation. A commonly used assumption is to write the incident field in the form (see for instance Liu 1970):

$$\mathbf{U}_i(\mathbf{x},t) = \varphi(t)\mathbf{V}_i(\mathbf{x},t) \quad (2)$$

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were $t \mapsto \varphi(t)$ is a deterministic continuous modulation function defined on 3^+; $\mathbf{V}_i(\mathbf{x},t)$ is a second order Gaussian non-homogeneous, mean square stationary centered stochastic field indexed on $\Gamma\times 3$ with values in 3^3 such that its auto-correlation function writes:

$$\mathbf{R}_{\mathbf{V}}(\mathbf{x},\mathbf{x}',t-t') = \int_{-\infty}^{+\infty} e^{i\omega(t-t')}\mathbf{S}_{\mathbf{V}}(\mathbf{x},\mathbf{x}',\omega)d\omega \tag{3}$$

with the transverse power spectral density $\mathbf{S}_{\mathbf{V}}$ defined on $\Gamma\times\Gamma\times 3$ with values in $\mathrm{Mat}_{\forall}(3,3)$. It has members for $m,n \in \{1,2,3\}$ two directions of the incident field:

$$S_{\mathbf{V},mn}(\mathbf{x},\mathbf{x}',\omega) = \sqrt{S_{0,m}(\omega)S_{0,n}(\omega)} \times \gamma_{mn}(\|\mathbf{x}-\mathbf{x}'\|,\omega) \qquad \mathbf{x},\mathbf{x}' \in \Gamma \tag{4}$$

where $S_{0,m}$ is the auto-spectrum of the incident field in direction m assumed independent of the position $\mathbf{x}$; γ_{mn} is the transverse coherency function defined on $3^+\times 3$ with values in $[0,1]$.

3.3 *Mechanical model and notations*

Let $\mathbf{u}(\mathbf{x},\omega)$ be the displacement field on Ω induced by the incident field $\mathbf{u}_i(\mathbf{x},\omega)$ in the frequency domain. Then it satisfies the homogeneous Navier equation in Ω:

$$\mathbf{Div}\boldsymbol{\sigma}(\mathbf{u}) + \rho\omega^2\mathbf{u} = \mathbf{0} \tag{5}$$

with traction-free conditions on $\Gamma_{as} \cup \Gamma_{ab}$ oriented by their outward normal unit vector $\mathbf{n}$:

$$\boldsymbol{\sigma}(\mathbf{u}) : \mathbf{n} = \mathbf{t}_{\mathbf{n}}(\mathbf{u}) = \mathbf{0} \tag{6}$$

$\boldsymbol{\varepsilon}(\mathbf{u})$ and $\boldsymbol{\sigma}(\mathbf{u})$ are the strain and stress tensors and $\mathbf{t}_{\mathbf{n}}(\mathbf{u})$ is the surface traction. ρ is the domain density, equal to ρ_b in the structure domain and ρ_s in the soil domain. In the following, the integral representations of interest for the structural displacement will be derived in the frequency domain written back in the time domain for the spectral analysis. The different fields introduced are written as functions of the circular frequency ω in 3 or the time t in 3^+ depending on the domain of analysis.

The notations below hold for two vector fields $\mathbf{u}$ and $\mathbf{v}$ on a given domain Ω_a with the Euclidean scalar product $\mathbf{u}.\mathbf{v}$:

$$\langle \mathbf{u},\mathbf{v}\rangle_{\Omega_a} = \int_{\Omega_a} \mathbf{u}(\mathbf{x}).\mathbf{v}(\mathbf{x})dV(\mathbf{x})$$

or their trace on a given surface Γ_a:

$$\langle \mathbf{u},\mathbf{v}\rangle_{\Gamma_a} = \int_{\Gamma_a} \mathbf{u}(\mathbf{x}).\mathbf{v}(\mathbf{x})dS(\mathbf{x})$$

For two tensors $\boldsymbol{\varepsilon}$ and $\boldsymbol{\sigma}$ we let:

$$\langle \boldsymbol{\varepsilon},\boldsymbol{\sigma}\rangle_{\Gamma_a} = \sum_{i,j}\int_{\Gamma_a} \varepsilon_{ij}(\mathbf{x}).\sigma_{ij}(\mathbf{x})dS(\mathbf{x})$$

4 INTEGRAL FORMULATION

The purpose of this paper is to introduce a formulation that allows engineers to compute directly relevant design quantities such as the structural displacements or accelerations at given significant points, including the effects of soil-structure interaction and free field spatial variability. We therefore introduce an integral representation of the displacement $\mathbf{u}(\boldsymbol{\xi}).\mathbf{e}$ at a point $\boldsymbol{\xi}$ on the structure along the direction of $\mathbf{e}$, $\|\mathbf{e}\| = 1$ when the overall domain Ω is impinged by an incident field $\mathbf{u}_i(\mathbf{x},\omega)$ as described above. It represents the displacement field in the soil in the vicinity of the foundation due to an earthquake when the structure has not been build yet.

This definition of the free field, which is supposed to come from infinity ($\Gamma_{s\infty}$), leads to decompose the displacement field $\mathbf{u}_s(\mathbf{x},\omega)$ in the soil domain into two parts, namely the incident free field and the diffracted field as (see for instance Aubry 86):

$$\mathbf{u}_s(\mathbf{x},\omega) = \mathbf{u}_i(\mathbf{x},\omega) + \mathbf{u}_d(\mathbf{x},\omega) \qquad \forall \mathbf{x} \in \Omega_s, \forall \omega \in 3 \tag{7}$$

The incident field and the diffracted field both satisfy the homogeneous Navier equation (5) in Ω_s and the traction-free surface condition on Γ_{as}. The diffracted field also satisfies the outgoing radiation conditions of the Sommerfeld type whereas the incident field does not.

4.1 *Direct integral representation*

Let $\mathbf{u}^g(\xi,\mathbf{x};\mathbf{e})$ be the Green function of the overall domain Ω when a unit load is applied at position ξ along direction $\mathbf{e}$. It is supposed to satisfy the heterogeneous Navier equation on Ω written in the frequency domain:

$$\mathbf{Div}\boldsymbol{\sigma}(\mathbf{u}^g) + \rho\omega^2\mathbf{u}^g + \delta(\mathbf{x}-\xi)\mathbf{e} = \mathbf{0} \tag{8}$$

The Green function also satisfies the traction-free surface condition on $\Gamma_{as} \cup \Gamma_{ab}$, and the outgoing Sommerfeld radiation conditions on $\Gamma_{s\infty}$. Hence we have the following integral equation on $\Gamma_{s\infty}$, that is written in the sense of a limit as $\|\mathbf{x} - \xi\| \to +\infty$:

$$\left\langle \mathbf{u}^g, \mathbf{t}_\mathbf{n}(\mathbf{u}_d) \right\rangle_{\Gamma_{s\infty}} - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u}_d \right\rangle_{\Gamma_{s\infty}} = 0 \tag{9}$$

A method to compute the Green function is given in section 4.

Now let us multiply Equation (5) by $\mathbf{u}^g$ and Equation (8) by $\mathbf{u}$, and then integrate over Ω. This gives us:

$$\begin{aligned} \mathbf{u}(\xi).\mathbf{e} &= \int_\Omega \mathbf{u}(\mathbf{x},\omega).\mathbf{e}\delta(\mathbf{x}-\xi)dV(\mathbf{x}) \\ &= \left\langle \mathbf{Div}\boldsymbol{\sigma}(\mathbf{u}), \mathbf{u}^g \right\rangle_\Omega - \left\langle \mathbf{Div}\boldsymbol{\sigma}(\mathbf{u}^g), \mathbf{u} \right\rangle_\Omega \end{aligned} \tag{10}$$

Integrating by parts, and then introducing the traction-free surface condition on $\Gamma_{as} \cup \Gamma_{ab}$, we get:

$$\mathbf{u}(\xi).\mathbf{e} = \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}), \mathbf{u}^g \right\rangle_{\Gamma_{s\infty}} - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u} \right\rangle_{\Gamma_{s\infty}} \tag{11}$$

By Equation (9) and the decomposition (7) of the displacement field in the soil domain, Equation (11) rewrites:

$$\mathbf{u}(\xi).\mathbf{e} = \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}_i), \mathbf{u}^g \right\rangle_{\Gamma_{s\infty}} - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u}_i \right\rangle_{\Gamma_{s\infty}} \tag{12}$$

The incident field and the Green function both satisfy the homogeneous Navier equation in the soil domain if ξ belongs to the structure domain. For such fields, Maxwell-Betti's reciprocity theorem for unbounded open domain leads to the following integral equation:

$$\left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}_i), \mathbf{u}^g \right\rangle_{\Gamma_{s\infty}} - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u}_i \right\rangle_{\Gamma_{s\infty}} + \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}_i), \mathbf{u}^g \right\rangle_\Gamma - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u}_i \right\rangle_\Gamma = 0 \tag{13}$$

where again it has been taken account of the traction-free boundary condition on $\Gamma_{as} \cup \Gamma_{ab}$,. We finally get the direct integral representation of the structural displacement at position ξ on the structure along the direction of $\mathbf{e}$ as:

$$\mathbf{u}(\xi).\mathbf{e} = \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}^g), \mathbf{u}_i \right\rangle_\Gamma - \left\langle \mathbf{t}_\mathbf{n}(\mathbf{u}_i), \mathbf{u}^g \right\rangle_\Gamma \tag{14}$$

4.2 *Indirect integral representation*

The direct integral representation of the structural displacement of interest requires the calculation of the traction field on the soil-structure interface Γ associated with the free field. In a stochastic approach based on the spectral analysis technique, it imposes the calculation of the auto-correlation function of the traction field $\mathbf{t_n}(\mathbf{U}_i)$ and the cross-correlation function of $\mathbf{U}_i$ and $\mathbf{t_n}(\mathbf{U}_i)$. This can be done formally by simple derivations, provided some regularity hypothesis for the auto-correlation function of $\mathbf{U}_i$. However we show here in this subsection that a much more efficient integral representation can be obtained, which does only depend on the free field on the interface. This approach is based on the explicit knowledge of the kernels of the single and double layer operators on Γ in the soil domain, these kernels being the first and second Green tensors $\mathbf{U}^G$ and $\mathbf{T}^G$ for the soil medium alone. If $(\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3)$ is an orthonormal basis of 3^3, then $[\mathbf{U}^G(\boldsymbol{\xi},\mathbf{x})]_{ij}$ and $[\mathbf{T}^G(\boldsymbol{\xi},\mathbf{x})]_{ij}$ are respectively the displacement and traction fields observed in the soil along $\mathbf{e}_j$ at the position $\mathbf{x}$ when a unit load is applied in it at $\boldsymbol{\xi}$ in the direction $\mathbf{e}_i$. They are computed analytically for homogeneous media, or numerically for stratified half-spaces.

The Green function $\mathbf{u}^g$ for the overall domain Ω satisfies the homogeneous Navier equation in the unbounded soil medium $\Omega_s\backslash\Omega_{sf}$ and the outgoing Sommerfeld radiation conditions, so that it is solution of the boundary integral equation:

$$\int_\Gamma \mathbf{U}^G(\boldsymbol{\xi},\mathbf{x})\mathbf{t_n}(\mathbf{u}^g)(\mathbf{x})dS(\mathbf{x}) - \int_\Gamma \mathbf{T}^G_{\mathbf{n}(\mathbf{x})}(\boldsymbol{\xi},\mathbf{x})\mathbf{u}^g(\mathbf{x})dS(\mathbf{x}) = \boldsymbol{\kappa}(\boldsymbol{\xi})\mathbf{u}^g(\boldsymbol{\xi}) \tag{15}$$

where:

$$\boldsymbol{\kappa}(\boldsymbol{\xi})\mathbf{u}^g(\boldsymbol{\xi}) = \lim_{\varepsilon\to 0}\int_{S(\boldsymbol{\xi},\varepsilon)\cap\Omega} \mathbf{T}^G_{\mathbf{n}(\mathbf{x})}(\boldsymbol{\xi},\mathbf{x})\mathbf{u}^g(\mathbf{x})dS(\mathbf{x}) \tag{16}$$

for $S(\boldsymbol{\xi},\omega)$ being the sphere centered at $\boldsymbol{\xi}$ with radius ε; $\mathbf{n}$ is the outward normal unit vector for $\partial(\Omega_s\backslash\Omega_{sf})$. The free field also satisfies the homogeneous Navier equation written in the bounded domain Ω_{sf} with outward normal unit vector $-\mathbf{n}$ on Γ. With the traction-free surface condition on Γ_{as} we get:

$$\int_\Gamma \mathbf{U}^G(\boldsymbol{\xi},\mathbf{x})\mathbf{t}_{-\mathbf{n}}(\mathbf{u}_i)(\mathbf{x})dS(\mathbf{x}) - \int_\Gamma \mathbf{T}^G_{-\mathbf{n}(\mathbf{x})}(\boldsymbol{\xi},\mathbf{x})\mathbf{u}_i(\mathbf{x})dS(\mathbf{x}) = \big(\mathbf{I} - \boldsymbol{\kappa}(\boldsymbol{\xi})\big)\mathbf{u}_i(\boldsymbol{\xi}) \tag{17}$$

Now let us consider the single layer elastic potential density $\mathbf{q}^g(\mathbf{x},\omega)$ associated with the displacement field $\mathbf{u}^g(\boldsymbol{\xi},\mathbf{x};\mathbf{e})$ and defined by:

$$\mathbf{u}^g(\boldsymbol{\xi}) = \int_\Gamma \mathbf{U}^G(\boldsymbol{\xi},\mathbf{x})\mathbf{q}^g(\mathbf{x})dS(\mathbf{x}) \tag{18}$$

The normal derivative of the single layer elastic potential is discontinuous through Γ and it is given by (Colton & Kress 1983):

$$\mathbf{t_n}(\mathbf{u}^g)(\boldsymbol{\xi}) = \int_\Gamma \mathbf{t}_{\mathbf{n}(\boldsymbol{\xi})}(\mathbf{U}^G)(\boldsymbol{\xi},\mathbf{x})\mathbf{q}^g(\mathbf{x})dS(\mathbf{x}) + \boldsymbol{\kappa}(\boldsymbol{\xi})\mathbf{q}^g(\boldsymbol{\xi}) \tag{19}$$

Multiplying the above equation by $\mathbf{u}_i$ and considering the algebraic properties of the integral operators with first and second Green tensors kernels, we get:

$$\begin{aligned}\left\langle \mathbf{t_n}(\mathbf{u}^g),\mathbf{u}_i\right\rangle_\Gamma &= \left\langle \mathbf{t}_{\mathbf{n}(\boldsymbol{\xi})}(\mathbf{U}^G)\mathbf{q}^g,\mathbf{u}_i\right\rangle_\Gamma + \left\langle \boldsymbol{\kappa}\mathbf{q}^g,\mathbf{u}_i\right\rangle_\Gamma \\ &= \left\langle \mathbf{q}^g,\mathbf{T}^G_{\mathbf{n}(\mathbf{x})}\mathbf{u}_i\right\rangle_\Gamma + \left\langle \mathbf{q}^g,\boldsymbol{\kappa}\mathbf{u}_i\right\rangle_\Gamma\end{aligned} \tag{20}$$

As we also have:

$$\left\langle \mathbf{t_n}(\mathbf{u}_i),\mathbf{u}^g\right\rangle_\Gamma = \left\langle \mathbf{t_n}(\mathbf{u}_i),\mathbf{U}^G\mathbf{q}^g\right\rangle_\Gamma = \left\langle \mathbf{U}^G\mathbf{t_n}(\mathbf{u}_i),\mathbf{q}^g\right\rangle_\Gamma, \tag{21}$$

we finally obtain from Equations (20) and (21):

$$\left\langle \mathbf{t_n}(\mathbf{u}^g), \mathbf{u}_i \right\rangle_\Gamma - \left\langle \mathbf{t_n}(\mathbf{u}_i), \mathbf{u}^g \right\rangle_\Gamma = \left\langle \mathbf{q}^g, \mathbf{T}^G_{\mathbf{n(x)}}\mathbf{u}_i - \mathbf{U}^G \mathbf{t_n}(\mathbf{u}_i) + \kappa \mathbf{u}_i \right\rangle_\Gamma \tag{22}$$

The indirect integral representation of the structural displacement at position ξ and along the direction of $\mathbf{e}$ is obtained injecting Equation (17) in Equation (22) and reads:

$$\mathbf{u}(\xi).\mathbf{e} = \left\langle \mathbf{q}^g, \mathbf{u}_i \right\rangle_\Gamma \tag{23}$$

We now show how to compute the Green function of the overall domain Ω. The single layer elastic potential density $\mathbf{q}^g$ is then calculated by the integral equation of the first kind (18).

Remark. This equation does not yield an unique solution for a set of discrete frequencies, called the "irregular frequencies", if the imaginary part of the wave number is zero. However Fredholm's alternative is verified for these frequencies, so that the numerical difficulties induced by them can be eliminated if a particular boundary integral equation is used (see Angelini & Hutin 1983 for an example in the acoustic case). We do not derive it here, but we will rather suppose that the imaginary part of the wave number is always positive, which is true when some damping is accounted for in the soil.

5 DERIVATION OF THE GREEN FUNCTION OF THE OVERALL DOMAIN

The evaluation of the Green function for the overall domain Ω is a classical soil-structure interaction problem for a unit load applied to the structure. Dynamic sub-structuring techniques are used in this respect. The theoretical framework is developed in Clouteau (1990). The elastodynamic problems for unbounded media are solved by regularized boundary integral equations (Aubry & Clouteau 1991). These methods have proved to be efficient for various problems such as soil-structure interaction, site effects or arch dam analysis under earthquake (see for instance Clouteau et al. 1995).

The equilibrium of the interface Γ is written in the weak sense in terms of the trace of $\mathbf{u}^g$ on Γ which is decomposed into N elementary fields, for instance rigid body motions or eigenmodes compatible with the foundation kinematic movement, by:

$$\mathbf{u}^g(\xi, \mathbf{x}; \mathbf{e}) = \sum_{p=1}^{N} c_p(\xi; \mathbf{e}) \boldsymbol{\psi}_p(\mathbf{x}) \qquad \forall \mathbf{x} \in \Gamma \tag{24}$$

The Green function is then written:

$$\mathbf{u}^g(\xi, \mathbf{x}; \mathbf{e}) = \sum_{p=1}^{N} c_p(\xi; \mathbf{e}) \boldsymbol{\psi}^*_p(\mathbf{x}) + \sum_{A=1}^{N_A} \alpha_A(\xi; \mathbf{e}) \boldsymbol{\phi}_A(\mathbf{x}) \qquad \forall \mathbf{x} \in \Omega_b \tag{25}$$

$$\mathbf{u}^g(\xi, \mathbf{x}; \mathbf{e}) = \sum_{p=1}^{N} c_p(\xi; \mathbf{e}) \mathbf{u}_{dp}(\mathbf{x}) \qquad \forall \mathbf{x} \in \Omega_s \tag{26}$$

where $\{\boldsymbol{\phi}_A, A = 1, \ldots N_A\}$ are the fixed basis eigenmodes of the structure with eigenvalues $\{\omega_A\}$ and critical dampings $\{\xi_A\}$, and $\{\mathbf{u}_{dp}, p = 1, \ldots N\}$ are the elementary scattered displacement fields on Ω_s when the boundary condition $\mathbf{u} = \boldsymbol{\psi}_p$ holds on Γ. They all satisfy the homogeneous Navier equation in the soil domain, as well as the outgoing Sommerfeld radiation condition on $\Gamma_{s\infty}$. $\{\boldsymbol{\psi}_p{*}\}$ are the extensions on the structure of the interface elementary fields. The virtual works principle applied to the structure for the test fields $\{\boldsymbol{\phi}_A\}$ and $\{\boldsymbol{\psi}_p\}$ respectively leads to the algebraic system:

$$\begin{bmatrix} \mathbf{K}^{\phi\phi} & -\omega^2 \mathbf{M}^{\psi\phi} \\ -\omega^2 \mathbf{M}^{\phi\psi} & \mathbf{K}^{\psi\psi} + \mathbf{K}_s \end{bmatrix} \begin{pmatrix} \boldsymbol{\alpha} \\ \mathbf{c} \end{pmatrix} = \begin{pmatrix} \mathbf{f}^\phi \\ \mathbf{f}^\psi \end{pmatrix} \tag{27}$$

where $\mathbf{f}^\phi$ and $\mathbf{f}^\psi$ are the projections of the unit load applied to the structure on the structural eigenmodes and interface modes respectively; $\mathbf{K}^{\phi\phi}$ and $\mathbf{K}^{\psi\psi}$ are the structure stiffness operator projections on its eigenmodes and interface modes respectively; $\mathbf{M}^{\phi\psi} = {}^T\mathbf{M}^{\psi\phi}$ are the participation factors of the interface modes on the structural eigenmodes with respect to the structure mass operator; at last, $\mathbf{K}_s$ is the symmetric foundation impedance given by:

$$K_{s,pq} = \left\langle \mathbf{t_n}(\mathbf{u}_{dp}), \boldsymbol{\psi}_q \right\rangle_\Gamma \tag{28}$$

The surface traction fields $\mathbf{t_n}(\mathbf{u}_{dp})$ are computed through a boundary integral equation on Γ written for each of the interface modes $\{\boldsymbol{\psi}_p\}$.

6 SPECTRAL ANALYSIS

The purpose of this section is to characterize the power spectral density (PSD) of the structural response when it is impinged by a random incident field described by its auto-correlation function / transverse power spectral density. A well known property of the PSD is that it allows to compute various design quantities of interest such as crossing rates or expected maxima. However it has to be kept in mind that the formula available to date to evaluate these quantities remain valid only for particular properties of the incident field.

The central theoretical point in spectral analysis of stochastic field is to identify the convolution filter applied to the excitation process, i.e. the free field in our case. It is given here by Equation (23) written back in the time domain:

$$\mathbf{u}(\boldsymbol{\xi},t).\mathbf{e} = \int_0^t \int_\Gamma \mathbf{q}^g(\mathbf{x}, t-\tau).\mathbf{u}_i(\mathbf{x},\tau)dS(\mathbf{x})d\tau \tag{29}$$

The design displacement $\mathbf{u}(\boldsymbol{\xi},t).\mathbf{e}$ is modeled by a stochastic process $U(t)$ indexed on 3^+ with values in 3. By linearity of (29), and given the properties of the incident stochastic field (see section 2), it is also Gaussian, zero-mean-valued and non-stationary. It is assumed second-ordered without any physical restriction to our model. The random design displacement $U(t)$ is therefore completely described by its auto-correlation function R_U defined on $3^+\times 3^+$ with values in 3. It is given straightforwardly by:

$$R_U(t,t') = \int_{-\infty}^{+\infty} d\omega \int_\Gamma \int_\Gamma \mathbf{T}^*(\mathbf{x},t,\omega)\mathbf{S}_\mathbf{V}(\mathbf{x},\mathbf{x}',\omega)\mathbf{T}(\mathbf{x}',t',\omega)dS(\mathbf{x})dS(\mathbf{x}') \tag{30}$$

with the following expression for a pseudo-equivalent deterministic force with values in $\forall^3$ and applied to the foundation:

$$\mathbf{T}(\mathbf{x},t,\omega) = \int_0^t e^{-i\omega\tau}\varphi(\tau)\mathbf{q}^g(\mathbf{x},t-\tau)d\tau \tag{31}$$

Equation (30) yields the following expression for the evolutionary power spectral density of the structural response:

$$S_U(t,\omega) = \int_\Gamma \int_\Gamma \mathbf{T}^*(\mathbf{x},t,\omega)\mathbf{S}_\mathbf{V}(\mathbf{x},\mathbf{x}',\omega)\mathbf{T}(\mathbf{x}',t,\omega)dS(\mathbf{x})dS(\mathbf{x}') \tag{32}$$

which is defined on $3^+\times 3$ with values in 3. It is easily shown that $S_U(t,\omega)$ is an even function of ω for all t in 3^+, as expected. The standard deviation of the structural response can be computed by:

$$\sigma_U^2(t) = 2\int_0^{+\infty} S_U(t,\omega)d\omega \tag{33}$$

7 APPLICATION

This analysis procedure is implemented for the Hualien mockup in Taiwan considering it rests on a rigid non-embedded circular spread footing of radius 5.41 m. The model characteristics are summarized in Tables 1 and 2 below. The mass density of the structure is 2.4 kg/m^3. The mass density of the soil is 2.42 kg/m^3, its viscous damping is 0.5% and its hysteretic damping is 2%. The modulation function is chosen as:

$$\begin{cases} \varphi(t) = (t/t_1)^2 \\ \varphi(t) = 1 \\ \varphi(t) = e^{-a(t-t_2)} \end{cases} \tag{34}$$

with a = 3.3, t_1 = 0.2s and t_2 = 1.0s. Of course this time window is not representative of a strong motion earthquake, however it has been chosen for convenience in our model, given its high stiffness and the band-broadness required by the frequency analysis. The transverse coherency function is taken as proposed by Luco & Wong (1986):

$$\gamma_{mn}(r,\omega) = \exp\left[-\left(\frac{\eta\omega r}{c}\right)^2\right] \tag{35}$$

for vertically incident seismic waves (no wave passage effect considered); c is the estimated wave celerity and η is a non-dimensional coherency parameter with values 0.0 through 0.5 in our computations.

Table 1. Hualien model characteristics – structure.

Structure	viscous damping (%)	frequency (Hz)
Horizontal flexure	2	10.8
Vertical compression	2	31.2

Table 2. Hualien model characteristics - soil

Soil	c_p (m/s)	c_s (m/s)	thickness (m)
Layer 1	1332	317	7
Layer 2	2001	476	∞

Figures 2 and 3 display the square-rooted normalized power spectral density function of the horizontal and vertical stationary displacements at the top of the structure for various values of the coherency parameter. The normalization coefficient is the square-rooted auto-spectrum of the incident stochastic field, which is supposed here independent of the direction. Figures 4 and 5 display the square-rooted normalized evolutionary power spectral density function of the horizontal and vertical displacements at the top of the structure for the coherency parameter η = 0.1. The normalization coefficient is again the square-rooted auto-spectrum of the incident stochastic field. It is observed that the time modulation has no effect on the obtained level of the PSD, except in the cases when the fundamental periods of the energetic peaks are much larger than the time window. For such peaks, their levels are reduced compared to the stationary displacements (see for instance Fig. 5).

We also compute the standard deviation of the structural responses. For that purpose, the auto-spectrum chosen for the stochastic incident field is the widely referenced Kanai-Tajimi spectrum. It is the power spectral density of the stationary acceleration of a 1 DOF oscillator with circular frequency ω_g and critical damping ξ_g, excited by a white noise of constant power s_0:

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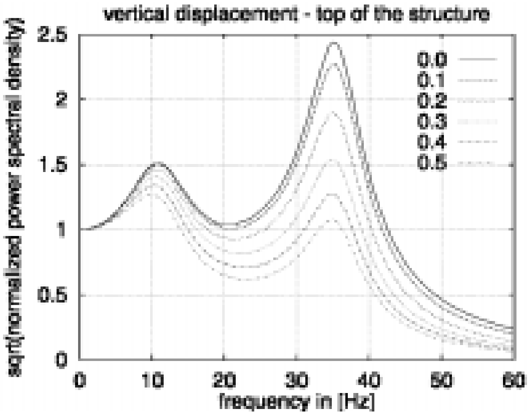

Figure 2. Square-rooted normalized power spectral density of the vertical displacement at the top of the structure for $\eta = 0.0 \sim 0.5$.

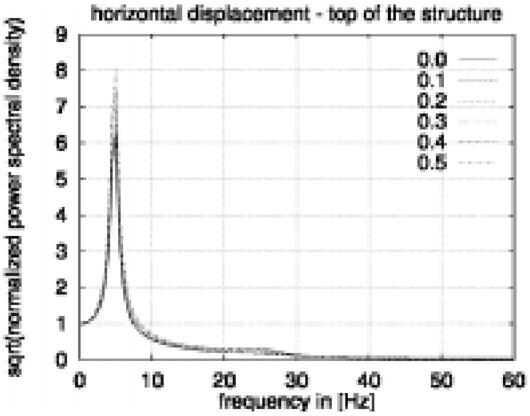

Figure 3. Square-rooted normalized power spectral density of the horizontal displacement at the top of the structure for $\eta = 0.0 \sim 0.5$.

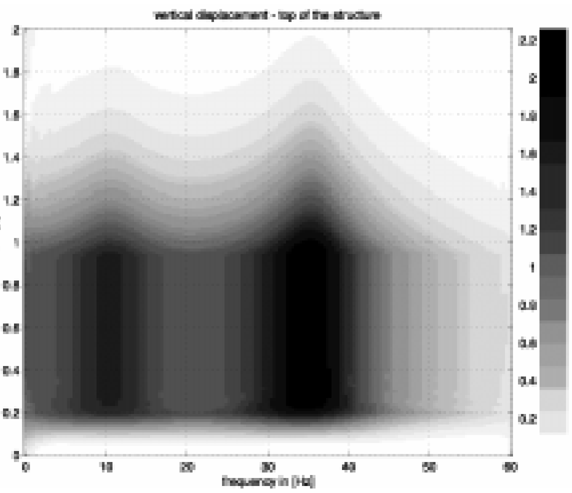

Figure 4. Square-rooted normalized evolutionary power spectral density of the vertical displacement at the top of the structure for $\eta = 0.1$.

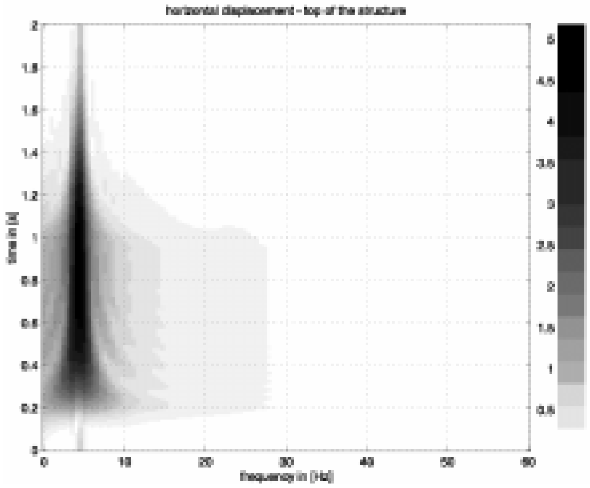

Figure 5. Square-rooted normalized evolutionary power spectral density of the horizontal displacement at the top of the structure for η = 0.1.

$$S_{KT}(\omega) = s_0 \frac{\omega_g^4 + 4\xi_g^2\omega_g^2\omega^2}{\left(\omega_g^2 - \omega^2\right)^2 + 4\xi_g^2\omega_g^2\omega^2} \tag{36}$$

This model has a singularity for $\omega = 0$ which is not consistent with recorded motions. To avoid this singularity, Clough & Penzien (1975) have proposed to filter the acceleration response process by another oscillator (ω_f,ξ_f). The incident field is then modeled by an acceleration process whose auto-spectrum is given by:

$$S_{CP}(\omega) = s_0 \left(\frac{\omega_g^4 + 4\xi_g^2\omega_g^2\omega^2}{\left(\omega_g^2 - \omega^2\right)^2 + 4\xi_g^2\omega_g^2\omega^2} \right) \times \left(\frac{\omega^4}{\left(\omega_f^2 - \omega^2\right)^2 + 4\xi_f^2\omega_f^2\omega^2} \right) \tag{37}$$

The standard deviations of the horizontal and vertical displacements at the top and bottom of the structure, normalized by $(s_0)^{1/2}$, are displayed on Figure 6 as functions of the time. The various parameters appearing in (37) are taken as proposed by Deodatis (1996). The trend of the modulation function is well reproduced. It is very similar for the vertical displacement at the top and bottom of the structure.

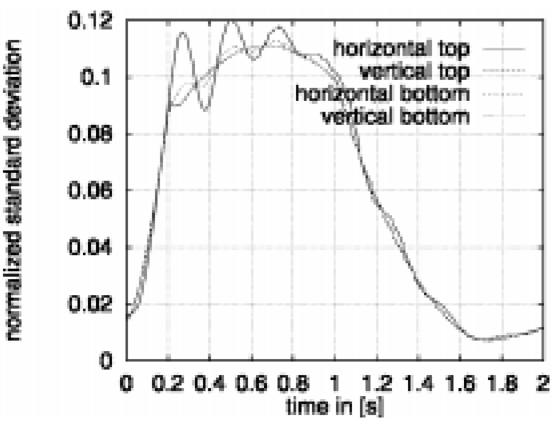

Figure 6. Normalized standard deviation of the horizontal and vertical displacements at the top and bottom of the structure for η = 0.1.

8 STOCHASTIC DECONVOLUTION

From a practical point of view, the statistical data available for the free field are limited to the ground surface only. The coherency function is most of the times empirically evaluated from records obtained at various locations. This means that the coherency functions for the incident displacement field in depth is rather unknown, although it is needed to perform a standard spectral analysis as described in the previous section for embedded foundations. We show here that for particular soil profiles and homogeneity assumptions for the surface motion a representation for the coherency function in depth is obtained. This method has been called stochastic deconvolution by Kausel & Païs (1987). We introduce it here for consistency purposes.

8.1 *Basic hypothesis*

The soil profile is supposed perfectly known and horizontally stratified. The incident seismic field is the superposition of plane waves propagating in all directions and it is supposed homogeneous on the ground surface, that is its transverse power spectral density writes:

$$\mathbf{S}_{\mathrm{V}}(\mathbf{x},\mathbf{x}',\omega)=\mathbf{S}_{\mathrm{V}}(\mathbf{x}-\mathbf{x}',\omega) \qquad \forall \mathbf{x},\mathbf{x}' \in \Gamma_{as} \tag{38}$$

Its power spectral measure has a density given by:

$$\hat{\mathbf{S}}_{\mathrm{V}}(\mathbf{k},\omega)=\int_{\Gamma_{as}} e^{-i\mathbf{k}.\mathbf{r}}\mathbf{S}_{\mathrm{V}}(\mathbf{r},\omega)d\mathbf{r} \tag{39}$$

where **k** is the horizontal wave vector. Finally it is supposed that no seismic source lies between the ground surface and the foundation highest depth.

Under these hypothesis, the displacement field propagating to any depth z can be evaluated by Thomson-Haskell's propagator method. Introducing the spatial Fourier transform in the horizontal plane as:

$$\hat{\mathbf{u}}(\mathbf{k},z,\omega)=\int_{\mathrm{R}^2} e^{-i\mathbf{k}.\mathbf{r}}\mathbf{u}(\mathbf{r},z,\omega)d\mathbf{r} \tag{40}$$

then the displacement field in the Fourier space at any depth z is given by:

$$\hat{\mathbf{u}}(\mathbf{k},z,\omega)=\mathbf{P}(\mathbf{k},z,\omega)\hat{\mathbf{u}}(\mathbf{k},0,\omega) \tag{41}$$

where the propagation operator **P** depends on the material characteristics of the soil layers between the ground surface and z.

8.2 *Transverse power spectral density in depth*

The transverse power spectral density of the free field in depth is obtained by:

$$\mathbf{S}_{\mathrm{V}}(\mathbf{x}_h-\mathbf{x}'_h,z,z',\omega)=\int_{\mathrm{R}^2} e^{i\mathbf{k}.(\mathbf{x}_h-\mathbf{x}_h')}\mathbf{P}(\mathbf{k},z,\omega)\hat{\mathbf{S}}_{\mathrm{V}}(\mathbf{k},\omega)\mathbf{P}^*(\mathbf{k},z',\omega)d\mathbf{k} \tag{42}$$

which shows that the incident stochastic field is not homogeneous in depth, although it is homogeneous on the ground surface. In the above $\mathbf{x}_h = \mathbf{x} - (\mathbf{x}.\mathbf{n})\mathbf{n}$. The transverse coherency matrix in depth is directly derived from (42) and is, for an auto-spectrum independent of the direction:

$$\gamma(\mathbf{r},z,z',\omega)=\frac{1}{S_0(\omega)}\int_{\mathrm{R}^2} e^{i\mathbf{k}.\mathbf{r}}\mathbf{P}(\mathbf{k},z,\omega)\hat{\mathbf{S}}_{\mathrm{V}}(\mathbf{k},\omega)\mathbf{P}^*(\mathbf{k},z',\omega)d\mathbf{k} \tag{43}$$

This expression can be used for the spectral analysis of section 5.

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9 CONCLUSION

An indirect integral representation of the structural displacement under earthquake has been presented that takes into account the soil-structure interaction effects. It is based on the knowledge of the fundamental solution (Green function) for the overall domain constituted by the structure and the soil when a unit load is applied at the location where the structural response is observed. This Green function is computed through classical sub-structuring techniques and boundary integral equations. The influence of the spatial variability of the incident field can be evaluated by a standard spectral analysis method. The approach has been implemented in the particular case of the Hualien containment model in Taiwan for a well known coherency function of the free field. The non-stationary effect is also taken into consideration by introducing a modulation function of the earthquake motion process. Finally, a method to compute its coherency function in depth is described in order to address the case of embedded foundations.

The proposed procedure can be improved by reducing the dimension of the frequency response matrix of the coupled system. This is achieved introducing a Karhunen-Loeve functional reduction of the incident free field which reduces its randomness dimension. The efficiency of the method is increased if the correlation distance of the free field increases. The formulation is given by Soize et al. (1986) in the acoustic case, and has been extended to the present case for a time modulated random excitation by Savin (1997).

ACKNOWLEDGEMENT

Financial support from Electricité de France EDF/DER is greatly acknowledged. Data for the Hualien containment model in Taiwan have been provided by EDF/SEPTEN.

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